

JAYKUMAR PRAJAPATI

MECHANICAL & INTEGRITY ENGINEER (EIT) · APEGA

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PROFESSIONAL SUMMARY

Mechanical & Integrity Engineer (EIT) registered with APEGA, specializing in **Finite Element Analysis (FEA)**, **Computational Fluid Dynamics (CFD)**, **API 579-1 / ASME FFS-1 Level 3 Fitness-for-Service**, and mechanical integrity evaluations of refinery pressure vessels, piping systems, and mechanical components. Master's graduate from the University of Calgary with proven success executing elastic-plastic stress simulations, transient thermal-fatigue analyses, and code-compliant design calculations (ASME BPVC VIII Div 2, B31.3, B31J, B16.5, PCC-1) registered under ABSA AB-520 regulatory requirements.

TECHNICAL COMPETENCIES & TOOLING

FEA & Structural Mechanics:	ANSYS Workbench & APDL, ANSYS Mechanical, ABAQUS, SolidWorks Simulation, Limit Load FEA, Thermal Transient & Fatigue Analysis, Random Vibration (PSD), Modal & Harmonic Resonance Analysis.
CFD & Fluid Mechanics:	ANSYS Fluent, Turbulence Modeling (SST k- ω , k- ϵ , Spalart-Allmaras), Grid Independence & Convergence, Aerodynamics, Aeroacoustics, Multi-phase Flow Dynamics.
CAD & BIM Softwares:	SolidWorks (Assemblies, Kinematics, GD&T), PTC CREO Parametric, AutoCAD (2D/3D Drafting, BOMs), Revit MEP (BIM Coordination), Bluebeam Revu (Markups & Takeoffs).
Codes, Standards & Regs:	API 579-1 / ASME FFS-1 (Level 3 FFS), ASME BPVC Section VIII (Div 1 & 2), ASME B31.3, ASME B31J (SIFs), ASME B16.5, ASME Sec II Part D, ASME PCC-1 App O, ABSA AB-520, ASHRAE 90.1, CSA Z662.
Programming & Data:	Python, MATLAB, Excel VBA (Macro Automation & Engineering Calculators), Power Query, QGIS (Geospatial Analysis).

PROFESSIONAL EXPERIENCE

Engineering Specialist — Mechanical Integrity | Zachry Integrity Engineering Ltd. May 2025 – Present | Calgary, AB

Zachry Integrity Engineering provides specialized mechanical integrity and engineering assessment services for refinery and petrochemical assets across North America.

- API 579-1 Level 3 FFS Assessments:** Supported comprehensive Level 3 Fitness-for-Service evaluations on critical refinery assets (vacuum towers, coke drums, fractionators, and cyclic piping systems) assessing local thin areas (LTAs), weld overlay distortions, and operating stress states.
- Finite Element Analysis (FEA):** Performed static, dynamic, and thermal transient fatigue analyses in **ANSYS Workbench and APDL** to evaluate localized stresses, plastic deformations, and low-cycle fatigue risks under severe thermal shock regimes.
- Custom SIF Assessments (ASME B31J):** Developed parametric FEA models for non-standard jacketed pipe cross configurations to derive custom Stress Intensification Factors (SIFs) and flexibility factors, achieving engineering registration under **ABSA AB-520**.
- Engineering Calculations & Review:** Evaluated client isometric drawings, ultrasonic inspection data, and thermography scans to produce calculation packages complying with ASME BPVC VIII Div 2, B31.3, B16.5, and PCC-1 Appendix O.
- Client & Field Collaboration:** Partnered with multidisciplinary engineering leads, field inspectors, and plant reliability personnel to deliver actionable repair strategies and equipment lifecycle recommendations.

Propulsion System Design Intern | Prime Toolings Sep 2024 – Jan 2025 | Remote

Prime Toolings specializes in precision tooling and engineering solutions for aerospace and industrial applications.

- Designed and optimized aerospace propulsion components and CAD layouts, balancing aerodynamic flow, structural rigidity, and manufacturability.
- Executed 30+ CFD simulations to evaluate boundary-layer airflow behavior, minimizing pressure drop across internal propulsion channels.
- Modeled high-stress turbine blade profiles, evaluating alloy selections and stress distributions under high rotational loading.
- Authored engineering calculation summaries, tolerance stack-ups, and simulation reports for internal engineering reviews.

Assistant Manager | Circle K Feb 2023 – Present | Calgary, AB

North America's leading convenience retail network. Supported high-volume daily retail operations and workforce leadership.

- Supervised store operations, inventory management, cash audits, and workplace health & safety compliance across teams.
- Engineered a custom **Excel VBA inventory tracking system** that automated stock audit calculations, significantly reducing stockout occurrences.
- Trained, scheduled, and led team members, fostering high customer satisfaction and continuous operational efficiency.

Aerospace Vehicle Design Project Trainee | Brahmastra Aerospace Nov 2022 – Jan 2023 | Remote

- Engineered high-lift aircraft wing profiles and internal structural spar/rib assemblies using **CATIA V5**.
- Conducted CFD simulations across multiple angles of attack (2°–16°) to evaluate boundary-layer separation and stall margins.
- Developed parameterized sounding rocket models and jet engine turbine blade profiles for aerodynamic stability simulations.

SELECTED ENGINEERING & INTEGRITY PROJECTS

PRESSURE EQUIPMENT, PIPING INTEGRITY & PROCESS SAFETY

Refinery Pressure Equipment Fitness-for-Service (API 579-1 / ASME FFS-1 Level 3) [ANSYS Mechanical](#) · [APDL](#) · [ASME Section VIII](#)

Constructed 3D non-linear FEA models of a Vacuum Tower and Coker Fractionator. Evaluated measured metal loss profiles, weld overlay distortions, and operating thermal loads to validate structural code compliance and establish safe remaining life margins.

Custom Stress Intensification Factors (SIFs) for Jacketed Crosses [ANSYS APDL](#) · [ASME B31J](#) · [ASME B31.3](#) · [ABSA AB-520](#)

Conducted FEA to compute custom fatigue-based SIFs and flexibility factors for non-standard jacketed cross configurations absent from ASME B31J tables. Authored formal engineering reports registered and approved by ABSA.

Coke Drum Bottom Flange Cyclic Thermal Transient FEA [ANSYS Workbench](#) · [ASME VIII Div 2](#) · [Thermal Transient](#)

Simulated severe thermal quench shock cycles using field thermography data. Conducted a parametric sensitivity study across 5", 6", and 7" flange geometries to mitigate low-cycle fatigue risks and eliminate bolting leak issues.

Reactor Catalyst Beds & Outlet Screen Limit Load Analysis [Limit Load FEA](#) · [ASME BPVC VIII](#) · [Structural Integrity](#)

Performed elastic-plastic limit load analysis on reactor internal catalyst support grids and 'elephant stool' collector under severe differential pressure (ΔP) loads, establishing safe operating threshold boundaries.

Anchor Bolt Dynamic & Resonance Analysis [Vibration Analysis](#) · [Modal Analysis](#) · [Structural Dynamics](#)

Processed vibration velocity spectra from rotating machinery foundations to evaluate dynamic stress amplitudes and natural resonance frequencies on anchor bolts, delivering torque retention and fatigue mitigation guidelines.

Alberta Pipeline Integrity & Proximity GIS Risk Model [QGIS](#) · [Spatial Buffering](#) · [NAD83 / UTM 12N](#) · [Asset Integrity](#)

Developed an automated geospatial risk analysis model in QGIS mapping Alberta's pipeline infrastructure relative to water bodies, highways, and high-consequence areas (HCAs) to prioritize pipeline direct assessment programs.

STRUCTURAL MECHANICS, CAD KINEMATICS & AERODYNAMICS

Structural Stress & Deflection Analysis of a Gantry Crane [SolidWorks FEA](#) · [ASTM A36](#) · [Structural Mechanics](#)

Modeled a structural gantry crane with W18×60 beams and HSS columns. Applied pin-roller boundary conditions, confirming maximum bending stress of 86.81 MPa ($FOS = 2.88$) and 6.84 mm mid-span deflection compliant with L/600 limits.

Random Vibration Analysis of Heavy-Duty Shock Absorber [ANSYS FEA](#) · [Power Spectral Density \(PSD 5–50 Hz\)](#)

Subjected 3D solid-element shock absorber assembly to random PSD acceleration profiles, mapping von Mises $1\sigma/2\sigma/3\sigma$ stress distributions to isolate fatigue vulnerability hot spots.

4-Cylinder Engine Assembly & Kinematic Simulation [SolidWorks](#) · [Dynamic Mates](#) · [Kinematics](#) · [GD&T](#)

Designed a fully constrained 4-cylinder engine assembly with dynamic piston/crankshaft mates, producing kinematic motion animations and ANSI Y14.5 GD&T manufacturing drawings.

Finite Span 3D Wing CFD & Mesh Convergence Analysis [ANSYS Fluent](#) · [Python](#) · [MATLAB](#) · [Prandtl Theory](#)

Conducted 3D fluid simulations and mesh convergence studies on a finite wing, resolving wingtip vortices and validating lift-curve slopes against classical Prandtl Lifting Line Theory.

Jet Engine Noise Reduction Analysis via Chevron Nozzle [ANSYS Fluent](#) · [Aeroacoustics](#) · [SolidWorks](#)

Simulated acoustic power level (dB) dissipation and turbulent kinetic energy fields of a sawtooth chevron nozzle, demonstrating enhanced shear-layer mixing that significantly attenuates low-frequency jet noise.

EDUCATION

Master of Engineering (M.Eng.) in Mechanical & Manufacturing Engineering Calgary, AB

[Schulich School of Engineering, University of Calgary](#) · Calgary, AB, Canada

Core Areas: Mechanical Integrity, Advanced Stress Analysis, Finite Element Methods (FEM), Manufacturing Optimization.

Bachelor of Engineering (B.Eng.) in Aeronautical Engineering Gujarat, India

[Gujarat Technological University](#) · Gujarat, India

Core Areas: Aerodynamics, Fluid Mechanics, Computational Fluid Dynamics (CFD), Propulsion & Aircraft Structures.

PROFESSIONAL CERTIFICATIONS & ACCREDITATIONS

Certificate in Advance Bluebeam — [PTS CAD EXPERT \(MSME India\)](#) (Jul 2026) Process Improvement Foundations — [LinkedIn Learning \(NASBA CPE\)](#) (Jul 2026)

Revit for Mechanical & Plumbing Design — [LinkedIn Learning](#) (Feb 2025) Lean Six Sigma Green Belt — [Continuous Improvement Org](#) (Feb 2025)

Reading HVAC, Plumbing & P&ID Drawings — [Drawing Certification](#) (Mar 2025) HVAC Engineering 101 Mastery Series — [Udemy](#) (Jan 2025)

Advanced PTC CREO Parametric Product Design — [CREO Academy](#) (Feb 2025) AutoCAD 3D Modeling for Mechanical Designs — [LinkedIn Learning](#) (Mar 2025)

Reading HVAC Drawings & Blueprints — [SkillCat \(IACET Accredited\)](#) (Apr 2025) Industrial Innovation through IIoT — [Mastering Up](#) (May 2025)

